# Jingze Dai

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## SKILLS AND CAPABILITIES (Front-End | Back-End | Full-Stack | DevOps | CI/CD | Machine Learning)

**Technical Skills**: **Advanced in** MongoDB, XML, OpenGL, OpenCV, AWS/GCP, DB2, Jenkins, Azure, GraphQL, Agile, R, SDLC, PyTorch/TensorFlow, Typescript, PostgreSQL, Docker, Golang; **Proficient in** Python, Java, C/C++, C#, Latex, .NET, Haskell, jQuery, Next.js, Spring, JavaScript, HTML, CSS, Ruby, Oracle, PostgreSQL/MySQL, Scala, JSON, Ansible, PowerBI, Kubernetes, SciPy/NumPy, React, Unity, Kotlin, git, Ubuntu, Linux/Unix, Android/MacOS, Swift, Rust, Xcode, MS Word, PowerShell, Excel, Scikit-Learn, Vue.js, TCP/IP, Selenium, Groovy, OpenAI (GPT), NoSQL, Apache (Hadoop & Spark), ETL.
**Certifications:** IBM Data Science, Machine Learning Specialization (Coursera), SQL **Advanced** & Angular, Rest APIs, Node.js **Intermediate** Certificate (HackerRank), Salesforce Development Professional, SAS **Advanced**, Deep Learning A-Z (Udemy).

## WORK EXPERIENCE

**Software Research Assistant** — Jan 2024 - Present
Cornell University & McMaster University — Ithaca, NY

- Wrote 6 conditions' stochastic energy systems' simulation models, and conducted independent proofs for each part, which determined the formulas of optimal parameters triggering the minimal two-layer cost. (https://arxiv.org/pdf/2507.03510)
- Devised 20 image-transformed CNN methods to detect V2X misbehaviour messages, reaching a 20-class detection accuracy of 98% and a detection time within 0.06 seconds, using explainable AI methods to select 3 most important driving features.
- Read 132 recent IoT publications and summarized a survey indicating recent XAI applications. (**Accepted** at **ISICN 2025**)

**Computer Science Course Teaching Assistant (CS 1XC3 & CS 3SH3)** — May 2023 - Dec 2023
McMaster University — Hamilton, ON

- Crafted 16 categories of operating system functions, taught C programming and LaTeX documentation to 200+ students, which achieved a 95% positive feedback rate for clarity and effectiveness.
- Evaluated students' 32 C code assignments and lab projects about multithreading and caching, ensuring fair grading and providing detailed feedback that clarified 110+ points of confusion for students.

**Software Developer (Co-op)** — Sept 2021 - Apr 2022
Government of Ontario, CYSSC services (Ministry of Children, Community, and Social Services) — Toronto, ON

- Developed 5 different MCCSS service websites, and embedded a server with user authentication and profile viewing APIs.
- Inserted login, questionnaire and appointment web applications, and constructed a MySQL 3-table user database.
- Constructed an Angular automated tester to test the whole website (Simulating human user logins and fingerprint checks).
- Analyzed 1000+ lines of user data, and summarized user appointment's actual childcare needs and time distributions.
- Designed 220+ pytest & JUnit test cases, which showed 18 errors and exceptions. Mitigated all the mentioned errors.

## EDUCATION

**Statistics, Cornell University** — Ithaca, NY
Master of Professional Studies (M.P.S) — Sept 2025 - Present
**Major in Statistics**, with specialization in **Data Science, GPA 3.9/4.0**
**Relevant Coursework**: Algorithms, Data Structures, Data Mining, Machine Learning, Artificial Intelligence, Natural Language Processing, Image Processing, Cloud/High-Performance Computing, Database Systems, Stochastic Calculus, Large Language Models, Time Series/Multivariate Analysis, Big Data Management, Robotics, Quantum Computing, Large-Language Models.

## PROJECTS

**MuseScore - Music Notation and Composition (https://musescore.com/)** — Apr 2023 – Present

- Contributed to MuseScore Google Summer of Code (GSoC) project, code at https://github.com/musescore/MuseScore.
- Implemented 3 statuses' cursor shape effects when dragging movable panels, and improved panels' merge/detach functions.
- Engineered 2 novel approaches to insert braille panels on the view menu, and designed 1 hotkey to enable/disable the braille panel. Inserted 2 optional arrows to increase the braille function visibility.

**Drasil - Generate All the Things (https://jacquescarette.github.io/Drasil/)** — May 2022 - Present

- Solved 30+ problems of the Drasil research project, supervised by Dr. Jacque Carette and Dr. Spencer Smith at McMaster University. Initial publication available at https://www.cas.mcmaster.ca/~carette/publications/DrasilPositionPaper.pdf.
- Mitigated 25 issues from Haskell automatic generations, improved 32 existing code segments with the expected format.
- Added 8 groups' references and environmental constraints to the SRS documentation, added 20+ components to its website.
- Inspected 7 tricky and complicated execution errors' messages, and discovered their reasons and fixing methods.